

Problem Set #10

57.4.10

1

$$L = \frac{m}{2} (\dot{x}^2 + \dot{y}^2) - a (x^2 + y^2)^{1/2}$$

$$x = \rho \cos \varphi$$

$$y = \rho \sin \varphi$$

$$\dot{x} = \dot{\rho} \cos \varphi - \rho \dot{\varphi} \sin \varphi$$

$$\dot{y} = \dot{\rho} \sin \varphi + \rho \dot{\varphi} \cos \varphi$$

$$L = \frac{m}{2} (\dot{\rho}^2 + \rho^2 \dot{\varphi}^2) - a \rho$$

$$p_\rho = \frac{\partial L}{\partial \dot{\rho}} = m \dot{\rho}$$

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m \rho^2 \dot{\varphi}$$

$$H(\rho, \varphi, p_\rho, p_\varphi) = -L + p_\varphi \dot{\varphi} + p_\rho \dot{\rho} \quad \Big| \quad \dot{q} = \dot{q}(q, p)$$

$$= \frac{1}{2m} p_\rho^2 + \frac{1}{2m} \frac{1}{\rho^2} p_\varphi^2 + a \rho$$

ii If $\langle \rho, \varphi | P_\rho | \Psi \rangle = \frac{\hbar}{i} \frac{\partial}{\partial \rho} \langle \rho, \varphi | \Psi \rangle,$

and

$$\int_0^\infty d\rho \int_0^{2\pi} \rho d\varphi |\rho, \varphi\rangle \langle \rho, \varphi| = 1$$

then, with $\psi(\rho, \varphi) \equiv \langle \rho, \varphi | \Psi \rangle,$

$$\langle \Psi_1 | P | \Psi_2 \rangle$$

$$= \int_0^\infty d\rho \int_0^{2\pi} \rho d\varphi \langle \Psi_1 | \rho, \varphi \rangle \frac{\hbar}{i} \frac{\partial}{\partial \rho} \langle \rho, \varphi | \Psi_2 \rangle$$

$$= \int_0^\infty d\rho \int_0^{2\pi} \rho d\varphi \psi_1^*(\rho, \varphi) \frac{\hbar}{i} \frac{\partial}{\partial \rho} \psi_2(\rho, \varphi)$$

$$= \int_0^\infty d\rho \int_0^{2\pi} \rho d\varphi \left[\left(\frac{\hbar}{i} \frac{\partial}{\partial \rho} \psi_1(\rho, \varphi) \right)^* \psi_2(\rho, \varphi) - \frac{\hbar}{i\rho} \psi_1^*(\rho, \varphi) \psi_2(\rho, \varphi) \right]$$

$$+ \int_0^{2\pi} d\varphi \frac{\hbar}{i} \left[\rho \psi_1^*(\rho, \varphi) \psi_2(\rho, \varphi) \right] \Big|_{\rho=0}^{\rho=\infty}$$

by integration by parts.

The surface term vanishes because the wavefunctions are finite at $\rho=0$ and vanish as $\rho \rightarrow \infty$ faster than $\rho^{-1/2}$.

Indeed, let $\psi \sim \rho^p$ as $\rho \rightarrow \infty$.

Then $\int_0^{2\pi} d\varphi \int_0^{\infty} \rho d\rho |\psi|^2 < \infty$ implies

$p < -1$. Hence the 2^d condition is satisfied.

And let $\psi \sim \rho^q$ as $\rho \rightarrow 0$.

$\int_0^{2\pi} d\varphi \int_0^{\infty} \rho d\rho |\psi|^2 < \infty$ only implies $q > -1$.

But we also need

$$\langle \psi | H | \psi \rangle < \infty.$$

Since

$$H = \frac{-\hbar^2}{2m} \frac{\partial^2}{\partial \rho^2} + \dots$$

this requires (provided some other mild assumption concerning the behaviour of V at $\rho \rightarrow 0$ are satisfied)

$$\int_0^{2\pi} d\varphi \int_0^{\infty} dp \, p \left| \frac{\partial \psi}{\partial p} \right|^2 < \infty$$

Hence $q > 0$ and the first condition is satisfied.

Thus

$$\begin{aligned} \langle \psi_1 | P | \psi_2 \rangle &= \langle \psi_2 | P | \psi_1 \rangle^* \\ &= \frac{\hbar}{i} \langle \psi_1 | \frac{1}{p} | \psi_2 \rangle \end{aligned}$$

which means that $P \neq P^\dagger$.

However, if

$$\langle p, \varphi | P_e | \psi \rangle = \frac{\hbar}{i} \left(\frac{\partial}{\partial p} + \frac{1}{2p} \right) \langle p, \varphi | \psi \rangle$$

then

$$\begin{aligned} \langle \psi_1 | P | \psi_2 \rangle &= \int_0^{\infty} p dp \int_0^{2\pi} d\varphi \, \psi_1^* \frac{\hbar}{i} \left(\frac{\partial}{\partial p} + \frac{1}{2p} \right) \psi_2 \\ &= \int_0^{\infty} p dp \int_0^{2\pi} d\varphi \, \psi_2 \left[\frac{\hbar}{i} \left(\frac{\partial}{\partial p} + \frac{1}{p} - \frac{1}{2p} \right) \psi_1 \right]^* \end{aligned}$$

$$\langle \psi_1 | P | \psi_2 \rangle = \langle \psi_2 | P | \psi_1 \rangle^*$$

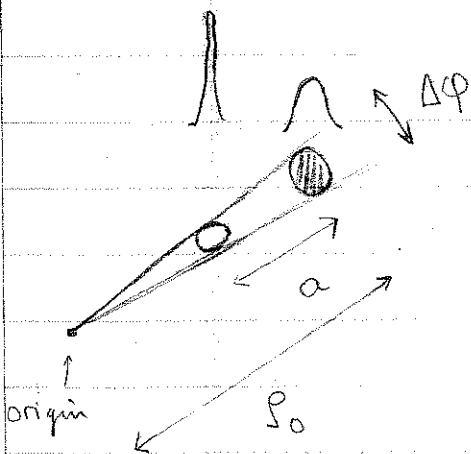
which means that $P = P^\dagger$.

iii. The situation is very different.

In Ex. 7.4.9. $f(x)$ must be real for P to be hermitean, whereas here

$$P_\varphi = \frac{\hbar}{i} \frac{\partial}{\partial \varphi} + \frac{\hbar}{i} \frac{1}{2\varphi}$$

pure imaginary



Consider a wavefunction $\psi(\rho, \varphi)$ localized in the shaded area.

Apply the unitary operator

$$U(a) = e^{+ia/\hbar P_\varphi}$$

to the corresponding state.

The new state is localized in the unshaded area, which is smaller than the shaded area. The new wavefunction must therefore have larger amplitude than the old one.

$$|\Psi'\rangle = U(a) |\Psi\rangle$$

$$\Psi'(r, \varphi) \approx \Psi(r+a, \varphi) \left(\frac{r+a}{r}\right)^{1/2}$$

for small a

to insure that

$$\int_0^{2\pi} d\varphi \int_0^{\infty} r dr |\Psi'(r, \varphi)|^2 = \int_0^{2\pi} d\varphi \int_0^{\infty} r dr |\Psi(r, \varphi)|^2 = 1$$

$$\Psi'(r, \varphi) \approx \left[\Psi(r, \varphi) + a \frac{\partial}{\partial r} \Psi(r, \varphi) \right] \left(1 + \frac{a}{2r}\right)$$

$$\approx \Psi(r, \varphi) + a \left(\frac{\partial}{\partial r} + \frac{1}{2r} \right) \Psi(r, \varphi)$$

Thus $\frac{i}{\hbar} P_r = \frac{\partial}{\partial r} + \frac{1}{2r}$

iv. $H = \frac{1}{2m} (P_r^2 + \frac{1}{r^2} P_\varphi^2) + a r$

$$\rightarrow \frac{-\hbar^2}{2m} \left[\left(\frac{\partial}{\partial r} + \frac{1}{2r} \right)^2 + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2} \right] + a r$$

$$H \rightarrow -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{2\rho} \frac{\partial}{\partial \rho} + \frac{\partial}{\partial \rho} \left(\frac{1}{2\rho} \cdot \right) + \frac{1}{4\rho^2} \cdot \right. \\ \left. + \frac{1}{\rho^2} \frac{\partial^2}{\partial \varphi^2} \cdot \right) + a\rho$$

$$= -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} - \frac{1}{2\rho^2} + \frac{1}{4\rho^2} \cdot \right. \\ \left. + \frac{1}{\rho^2} \frac{\partial^2}{\partial \varphi^2} \cdot \right) + a\rho$$

$$H \rightarrow -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} - \frac{1}{4\rho^2} + \frac{1}{\rho^2} \frac{\partial^2}{\partial \varphi^2} \right) + a\rho$$

S7.5.4

$$\begin{aligned} \underline{a.} \quad -\frac{\partial}{\partial \beta} \ln Z &= \frac{-1}{Z} \frac{\partial}{\partial \beta} \left(\sum_i e^{-\beta E_i} \right) \\ &= \sum_i E_i e^{-\beta E_i} \frac{1}{Z} = \sum_i E_i P_i = \bar{E} \end{aligned}$$

$$\begin{aligned} \underline{b.} \quad Z_d &= \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dp e^{-\beta \left(\frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 \right)} \\ &= \sqrt{\pi \frac{2m}{\beta}} \cdot \sqrt{\pi \frac{1}{\beta m \omega^2}} = \frac{2\pi}{\omega \beta} \end{aligned}$$

$$\bar{E}_d = -\frac{\partial}{\partial \beta} \ln Z_d = \frac{1}{\beta} = kT$$

$$\underline{c.} \quad Z_{qu} = \sum_{n=0}^{\infty} e^{-\beta \left(n + \frac{1}{2} \right) \hbar \omega}$$

$$= e^{-\frac{1}{2} \beta \hbar \omega} \sum_{n=0}^{\infty} \left(e^{-\beta \hbar \omega} \right)^n$$

$$= \frac{e^{-\frac{1}{2} \beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}} = \frac{1}{2 \sinh \left(\frac{\beta \hbar \omega}{2} \right)}$$

$$\bar{E}_{qu} = \frac{-\partial}{\partial \beta} \left[\ln(e^{-\frac{1}{2}\beta \hbar \omega}) - \ln(1 - e^{-\beta \hbar \omega}) \right]$$

$$= \hbar \omega \left[\frac{1}{2} + \frac{e^{-\beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}} \right]$$

$$= \hbar \omega \left[\frac{1}{2} + \frac{1}{e^{\beta \hbar \omega} - 1} \right]$$

d. For $\beta \hbar \omega \ll 1$, or $kT \gg \hbar \omega$,

$$\bar{E}_{qu} \approx \hbar \omega \left(\frac{1}{2} + \frac{1}{\beta \hbar \omega + \frac{1}{2}(\beta \hbar \omega)^2 + O(\beta \hbar \omega)^3} \right)$$

$$\approx \hbar \omega \left(\frac{1}{2} + \frac{1}{\beta \hbar \omega} \left(1 - \frac{1}{2}(\beta \hbar \omega) + O(\beta \hbar \omega)^2 \right) \right)$$

$$= kT (1 + O(\beta \hbar \omega)^2)$$

$$\underline{e} \quad \bar{E}_{clcr} = 3N_0 \quad \bar{E}_{cl} = 3N_0 kT$$

$$C_{cl}(T) = \frac{1}{N_0} \frac{\partial \bar{E}_{clcr}}{\partial T} = 3k$$

$$\bar{E}_{qucr} = 3N_0 \quad \bar{E}_{qu} = 3N_0 \hbar \omega \left(\frac{1}{2} + \frac{1}{e^{\beta \hbar \omega} - 1} \right)$$

$$C_{qu}(T) = \frac{1}{N_0} \frac{\partial \bar{E}_{qucr}}{\partial T} = \frac{1}{N_0} \left(\frac{d\beta}{dT} \right) \frac{\partial \bar{E}_{qucr}}{\partial \beta}$$

$$= \frac{1}{N_0} \left(\frac{-1}{kT^2} \right) 3N_0 \hbar \omega \frac{-\hbar \omega e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

$$= 3k \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2} \quad \text{with } \theta_E \equiv \frac{\hbar \omega}{k}$$

$$C_{qu}(T) \rightarrow 3k \left(\frac{\theta_E}{T} \right)^2 \frac{1}{\left(\frac{\theta_E}{T} \right)^2} = 3k \quad \text{for } T \gg \theta_E$$

$$\rightarrow 3k \left(\frac{\theta_E}{T} \right)^2 e^{-\theta_E/T} \quad \text{for } T \ll \theta_E$$

S8.6.1

$$V(x) = -fx$$

$$m \frac{d^2 x_d}{dt^2} = +f$$

$$\therefore x_d(t) = x_d(0) + \dot{x}_d(0)t + \frac{f}{2m}t^2,$$

$$\text{For } x_d(t') = x', \quad x_d(t'') = x''$$

$$x_d(t) = x' + v_0(t-t') + \frac{f}{2m}(t-t')^2$$

with

$$x'' = x' + v_0(t''-t') + \frac{f}{2m}(t''-t')^2.$$

Hence

$$v_0 = \frac{x'' - x'}{t'' - t'} - \frac{f}{2m}(t'' - t').$$

$$S_d = \int_{t'}^{t''} dt \left[\frac{m}{2} v^2 + f x \right]$$

$$= \int_{t'}^{t''} dt \left\{ \frac{m}{2} \left[v_0 + \frac{f}{m}(t-t') \right]^2 + f \left[x' + v_0(t-t') + \frac{f}{2m}(t-t')^2 \right] \right\}$$

$$S_{cl} = \int_{t'}^{t''} dt \left\{ \frac{m}{2} v_0^2 + f x' + 2 f v_0 (t-t') + \frac{f^2}{m} (t-t')^2 \right\}$$

$$= \left(\frac{m}{2} v_0^2 + f x' \right) (t''-t') + f v_0 (t''-t')^2 + \frac{f^2}{3m} (t''-t')^3$$

$$\frac{m}{2} v_0^2 (t''-t') + f v_0 (t''-t')^2$$

$$= \frac{m}{2} \left[v_0 (t''-t') + \frac{f}{m} (t''-t')^2 \right]^2 \frac{1}{t''-t'}$$

$$- \frac{f^2}{2m} (t''-t')^3$$

$$= \frac{m}{2} \left[x'' - x' + \frac{f}{2m} (t''-t')^2 \right]^2 \frac{1}{t''-t'}$$

$$- \frac{f^2}{2m} (t''-t')^3$$

$$= \frac{m}{2} \frac{(x''-x')^2}{t''-t'} + \frac{f}{2} (x''-x') (t''-t') - \frac{3}{8} \frac{f^2}{m} (t''-t')^3$$

$$\left(\frac{1}{8} - \frac{1}{2} = \frac{1-4}{8} = -\frac{3}{8} \right)$$

$$S_d = \frac{m}{2} \frac{(x'' - x')^2}{t'' - t'} + \frac{f}{2} (x'' + x')(t'' - t')$$

$$- \frac{1}{24} \frac{f^2}{m} (t'' - t')^3$$

$$\left(\frac{-3}{8} + \frac{1}{3} = \frac{-9+8}{24} = \frac{-1}{24} \right)$$

Therefore

$$U(x, t; x', 0) = A(t) \exp \left\{ \frac{i}{\hbar} \left[\frac{m}{2} \frac{(x-x')^2}{t} + \frac{f}{2} (x+x')t - \frac{f^2}{24m} t^3 \right] \right\}$$

See section 8.6

In particular, see Eq. (8.6.11)

$A(t)$ does not depend upon f .

$$\text{For } f=0, \quad U(x, t; x', 0) = \left(\frac{m}{2\pi \hbar i t} \right)^{3/2} e^{\frac{im}{2\hbar} \frac{(x-x')^2}{t}}$$

from Eq. (5.1.10)

Therefore, for $f \neq 0$,

$$A(t) = \left(\frac{m}{2\pi \hbar i t} \right)^{3/2} \quad \text{as well.}$$